

The SPO portfolio tree: model v1

The tree in plain words; the planted-threshold experiment; equations in the appendix

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Outline

1. What we do, and how we test it
2. How the tree learns, in plain words
3. The planted-threshold experiment (E2–E11)
4. A fair assessment, and what comes next

What we do, and how we test it

Problem

- weights of n assets every h days, from features x_t , with fees
- a decision problem, not a forecast

Model

- small tree $\rightarrow$ leaf scores $\rightarrow$ mean–variance optimizer with fees
- loss: decision regret through the optimizer
- splits: confirmed on the tree's own traded path, pruned on held-out weeks

Test

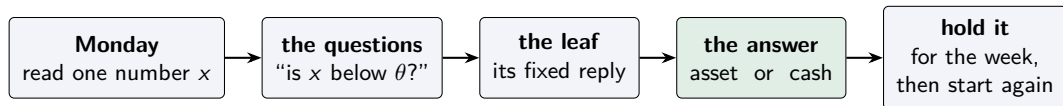
- synthetic market, one feature x , planted threshold d
- benchmark: *ideal rule* = optimizer fed $\mathbb{E}[r \mid x]$
- constant volatility; asset vs cash; weekly unless stated

Model v1

- frozen 24 September 2026
- 20,000+ fitted trees behind the numbers; validated on 100 datasets

Equations: appendix.

Every Monday: one number in, one decision out

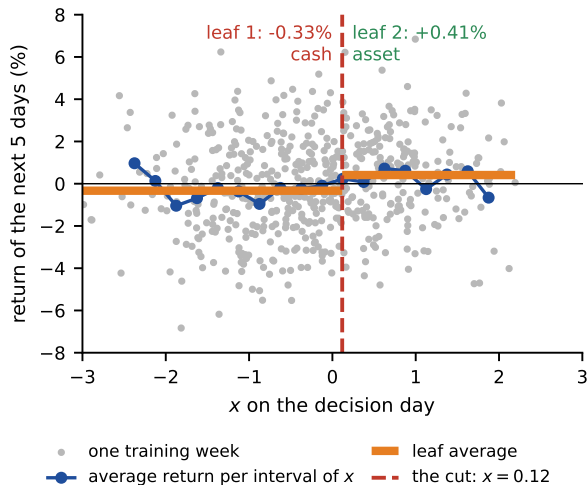


- The tree = the questions + one answer per leaf. Nothing else. Depth 3: at most 3 questions in a row, 8 leaves.
- **Learning the tree** = choosing where to cut (θ) and what each leaf answers.
- In the test: x = the planted feature, d = the planted threshold.

Simplification used in this section

No fees, no risk term: answer = all asset or all cash. Model v1 has both, but at $\lambda = 1$ and 3 bp its positions are 0 or 1 anyway: the pictures stay faithful.

One cut: sort the weeks by x , look at their returns



- Grey dot = one training week: x on the Monday, return of the week after. 567 weeks = the first 11 of the 15 training years; the last 4 are kept for the check.
- Blue: average return per interval of x (width 0.25). Left of the cut: losses; right: gains.
- Leaf 1: **cash**. Leaf 2: **asset**.
- The whole model = one number, the cut, learnt from these weeks. True cut of this market: $d = 0$.

One synthetic dataset; drift $\pm 25\%$ a year, volatility 16% (regime Sharpe 1.6); weekly; model v1.

A cut is judged by the money lost on the wrong weeks

A toy training set of 8 weeks

week	x	return	no cut	cut at 0
1	-1.6	-2.0%	asset (2.0)	cash
2	-0.9	-1.0%	asset (1.0)	cash
3	-0.5	+1.0%	asset	cash (1.0)
4	-0.2	-1.5%	asset (1.5)	cash
5	+0.3	+2.0%	asset	asset
6	+0.6	-0.5%	asset (0.5)	asset (0.5)
7	+1.1	+1.5%	asset	asset
8	+1.8	+2.0%	asset	asset
total return of the leaf (leaves)			+1.5%	-3.5, +5.0%
cost = money lost on the wrong weeks			5.0%	1.5%

in brackets: the money lost by that answer

The rule for one leaf

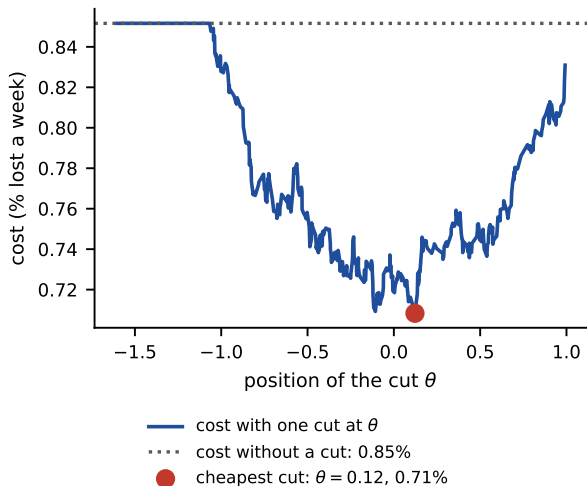
- answer: **asset** if its weeks gain in total, **cash** otherwise
- cost: money lost on the weeks where that answer is wrong: losses taken in the asset, gains missed in cash

Read the table

- no cut: 8 weeks gain +1.5% together $\Rightarrow$ asset every week; wrong on weeks 1, 2, 4, 6: cost 5.0%
- cut at 0: left leaf -3.5% $\Rightarrow$ cash, misses week 3; right leaf +5.0% $\Rightarrow$ asset, loses week 6: cost 1.5%
- cost 5.0% $\rightarrow$ 1.5%: the cut is worth making

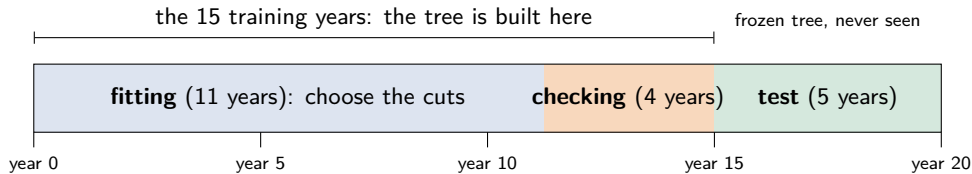
Real model: this cost = the *decision regret*, hindsight decision minus the leaf's decision, week by week.

Try every cut, keep the cheapest, then check it by trading it



- About 450 possible cuts between the sorted weeks; each leaf keeps at least 57 weeks, 10% of 567. Each cut scored by the cost of its two leaves; cheapest wins: the red dot.
- **Then the check by trading.** The rule is traded Monday after Monday over the 11 fitting years, from the money actually held. Cut kept only if the **Sharpe ratio** rises.
- v1 shortcut: the 9 deciles first, then every cut between the two deciles around the best: 8 times fewer candidates, same result.

The training loop, and the three periods



1. **Start.** One leaf, no cut: asset or cash for every week, whichever costs less.
2. **Grow.** The cut that lowers the cost the most (previous frame); kept only if trading it over the fitting years raises the Sharpe ratio.
3. **Repeat.** Best cut inside every leaf; the one that lowers the total cost the most is tried, kept if the Sharpe check passes. Stop when no candidate passes, or at 3 levels of cuts (8 leaves). Leaf ≥ 20 weeks and 10% of its parent.
4. **Check.** The last 4 training years, never used above. Every cut judged there, the tree traded with and without it: removed unless the cost falls and the Sharpe ratio rises; below the root, cost reduction ≥ 2 standard errors.
5. **Freeze.** Answer of each remaining leaf recomputed on all 15 training years. Nothing changes after this.
6. **Test.** The frozen tree traded on 5 further years it has never seen: the reported number.

What the real model adds back

In this section	In model v1	Why it matters
answer: asset or cash	a predicted return per leaf (e.g. +0.4% a week) → small optimizer: gain — fees — risk	at $\lambda = 1$ and 3 bp: still 0 or 1; sizing from $\lambda \approx 5$
no fees	3 bp per trade on the asset, inside the optimizer	in the fit: frozen holdings, not the fee the strategy pays (next frame)
no risk term	$\lambda w^2 \sigma^2$ with $\lambda = 1$	invisible at $\lambda = 1$; gradual positions at $\lambda = 5-10$
every cut tried	9 deciles, then every cut around the best; leaf ≥ 20 weeks and 10% of its parent	fewer candidates, steadier cuts, same Sharpe
plain check on the checking years	below the root: cost reduction ≥ 2 standard errors	removes the spurious second-level leaves
cost = money lost	decision regret through the optimizer, week by week	identical when positions are 0 or 1

The fee inside the fit is not the fee the strategy pays

During the fit of a leaf: frozen holdings

Optimizer run week by week from the holdings of the *current* tree's path, fixed for the round. Its fee penalty prices the move from those holdings to the candidate decision. **Not** the fees the strategy being built would pay: that strategy reaches each week with its own holdings.

During the replay: holdings updated

The strategy is traded week after week: holdings updated from its own previous decision, drifted by the returns; its fees counted. **Real fees: known only after the replay.** The replay is where a strategy is measured: Sharpe check of a cut, pruning check, test.

Side effect: a leaf score within 3 bp of the score that justifies the current holdings leaves the optimizer where it is (the freeze of E10). v2: no fee inside the fit; fees only where a strategy is really traded.

The base and model v1: the same training, three options switched on

	the base (every option off)	model v1 (frozen 24 September)
depth	at most 2 cuts in a row: 4 leaves	at most 3: 8 leaves
candidate cuts	every position between the sorted weeks	9 deciles, then every cut between the two deciles around the best; kept if it wins by 1 s.e.
leaf minimum	20 weeks	20 weeks and 10% of the parent leaf
check on the last 4 training years	cut stays if the cost falls and the Sharpe ratio rises there	same, plus cost reduction ≥ 2 s.e. below the root
same for both	$\lambda = 1$, fee 3 bp, weekly, 180-day covariance, cash as second asset, soft splits off; 15 training years (11 + 4), 5 test years; growth rule: largest cost reduction first, kept if the traded Sharpe rises	

Base: E2 to E8 (the first results below); v1: E9 to E11. E9, same 100 datasets: v1 = same Sharpe, smaller and cleaner trees. Full v1 table with reasons: Appendix B.

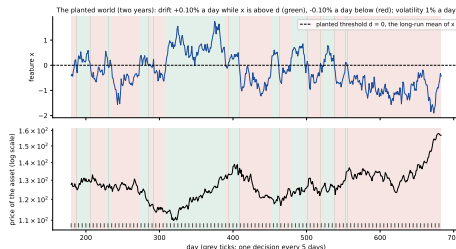
The test-bed: a market with a planted answer (constant volatility)

$$dx_t = k(m - x_t) dt + s\sqrt{2k} dW_t$$

$$\frac{dS_t}{S_t} = \mu_t dt + \sigma dB_t, \quad \mu_t = \begin{cases} +e\sigma & x_t \geq d \\ -e\sigma & x_t < d \end{cases}$$

t in trading days; W , B independent; x mean-reverting, stationary variance s^2 ; simulated daily with the exact one-day step

σ (daily)	1% (16% a year), both regimes
$e = \mu/\sigma$ (daily)	0.10, 0.05, 0.02: drift ± 25 , ± 13 , $\pm 5\%$ a year; regime Sharpe $e\sqrt{252} = 1.6, 0.8, 0.3$
k (daily)	0.02: half-life $\ln 2/k = 35$ days; 0.01 to 0.10 (69 to 7 days)
m, s^2, d	0, 1, 0: x standardized (mean 0, sd 1), threshold at its mean
h (trading days)	1, 2, 5, 10, 21: daily to monthly
data	daily; 15 years of training (3,780 days), 5 of test (1,260)

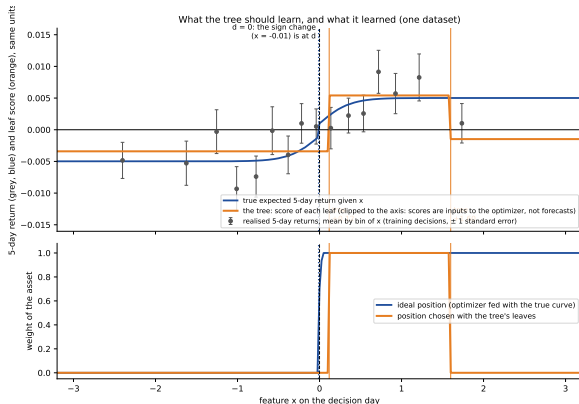


Measured on 40–100 datasets per cell

- first threshold vs planted $d = 0$
- test Sharpe: tree vs ideal rule ($\mathbb{E}[r | x]$ fed to the optimizer), rule at d , buy & hold
- share of the ideal gain captured

Second market, volatility threshold too: E11 at the end.

One fitted tree (weekly, edge 0.10)



What is drawn (x on the decision day)

- **blue**: true expected 5-day return given x ;
grey: realised returns, averaged by interval of x
- **orange**: the tree's score, one value per leaf
- **bottom**: the position each one implies

Key points

- **The root split lands on the planted threshold.**
- **Scores are not forecasts**: far outside the optimizer's window; positions match the ideal rule wherever the sign is right.
- **Far-right leaf: spurious**. A tail of x sent to cash; $v1$'s pruning hurdle removes such leaves.
- **At $\lambda = 1$ the ideal rule is a step too**. The threshold is the whole problem.

Results: the threshold found, the Sharpe earned (E2, base pipeline)

edge	h	split on x (%)	split threshold	annualized test Sharpe ratio				gain
			within 0.25	tree	ideal	rule at d	buy & hold	captured
0.10	1	97	95%	0.80	1.03	1.04	-0.01	76%
	2	99	92%	0.86	0.97	0.98	-0.01	85%
	5	94	81%	0.73	0.88	0.88	-0.01	79%
	10	89	70%	0.63	0.78	0.78	-0.01	78%
	21	81	60%	0.54	0.68	0.66	-0.01	75%
0.05	1	82	60%	0.12	0.49	0.49	-0.02	25%
	2	75	64%	0.21	0.46	0.46	-0.02	44%
	5	78	53%	0.24	0.43	0.42	-0.02	51%
	10	73	47%	0.17	0.39	0.38	-0.02	43%
	21	62	34%	0.17	0.35	0.34	-0.03	43%
0.02	5	54	26%	-0.03	0.15	0.13	-0.03	-3%

100 datasets per row (50 at edge 0.02). Edge 0.10, 0.05, 0.02 = drift ± 25 , ± 13 , $\pm 5\%$ a year; h in trading days. Split on x : trees keeping a split. Within 0.25: first threshold within 0.25 of the planted 0, among those. Sharpe: test years, mean over datasets, s.e. ≈ 0.04 . Ideal: optimizer fed $\mathbb{E}[r \mid x]$. Rule at d : asset iff $x \geq 0$, the threshold known. Buy & hold: always the asset. Gain: tree's excess return over the no-split tree, as a share of the ideal rule's. Fees $\approx 0.2\%$ a year weekly.

- **$\pm 25\%/yr$: found at every rate.** Within 0.25: 95% daily, 60% monthly; 75–85% of the ideal gain.
- **$\pm 13\%/yr$: graceful.** 62–82% split; 25–51% of a small gain (ideal 0.35–0.49).
- **$\pm 5\%/yr$: nothing to find in 15 years.** Half no split, half at random; ideal rule 0.15. Finding nothing = the right answer.
- **The threshold is the whole job.** Rule at d = ideal rule. Lost: threshold error + a few spurious leaves.
- **No signal, no gain.** Buy & hold earns zero: up in one regime, down in the other. All the gain is in the switch.

Decision rate against the speed of the feature (edge 0.10, 50 datasets per cell)

half-life of x	daily		weekly		monthly	
	tree	ideal	tree	ideal	tree	ideal
69 days ($k = 0.01$)	0.63	1.09	0.84	0.97	0.68	0.84
35 days ($k = 0.02$)	0.80	1.03	0.73	0.88	0.54	0.68
14 days ($k = 0.05$)	0.69	0.98	0.65	0.75	0.30	0.49
7 days ($k = 0.10$)	0.77	0.96	0.45	0.60	0.18	0.34

test Sharpe, tree and ideal rule

- daily $\approx$ weekly: more decisions = more fees, not more information
- monthly loses, more for a fast feature: half-life 7 d, 0.18 vs 0.34
- slow feature + daily decisions: over-trading (0.63 vs 1.09)
- $\Rightarrow$ weekly in v1

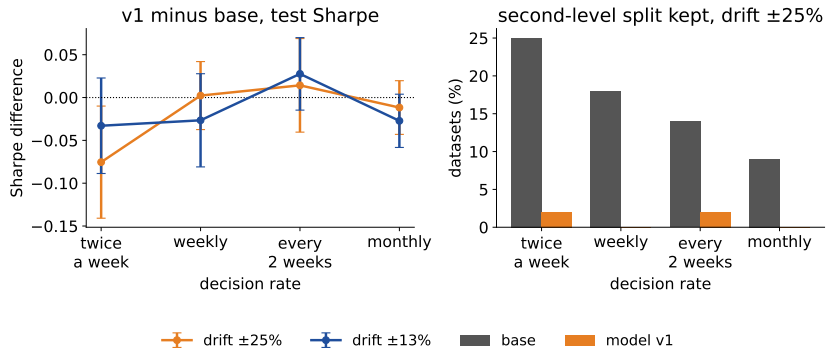
What the design options bought (E3–E5, E8; 40 datasets per cell, edge 0.10)

setting	annual. Sharpe, tree	
	weekly	2-day
base (every option off)	0.80	0.87
depth 3	0.80	0.89
$\lambda = 5$ / $\lambda = 10$	0.70 / 0.65	0.81 / 0.71
fee 10 bp	0.60	0.53
deciles + refinement	0.82	0.82
soft splits $\kappa = 1$	0.53	0.35
soft splits $\kappa = 0.25$	0.73	0.61
hurdle 2 s.e., every node	0.43	0.47
hurdle 2 s.e., below the root only	0.77	0.74
deciles + refin. + hurdle below root	0.80	0.79

base: every option off (depth ≤ 2 , every threshold screened, no pruning hurdle, $\lambda = 1, 3$ bp). Each row: one change. Ideal rule 0.91 weekly, 1.00 at 2 days; s.e. ≈ 0.04 ; last row = v1 at depth 2.

- **No option beats the base beyond noise.** All within 2 s.e.; what remains is the noise of the threshold itself.
- **Depth, λ , fees: as expected.** Depth 3: nothing, one threshold = one split. $\lambda = 5$ –10: gradual positions, cost 0.1–0.15. 10 bp: the fast rates stop trading.
- **Deciles + refinement: kept.** Same Sharpe, 9 candidates instead of hundreds, steadier thresholds.
- **Hurdle at every node: not kept.** Removes true root splits: 4 checking years cannot certify a split at 2 s.e.
- **Hurdle below the root only: kept.** Spurious second-level leaves: 25% $\rightarrow$ 5% at 2 days, none from weekly on; no Sharpe cost from weekly on.

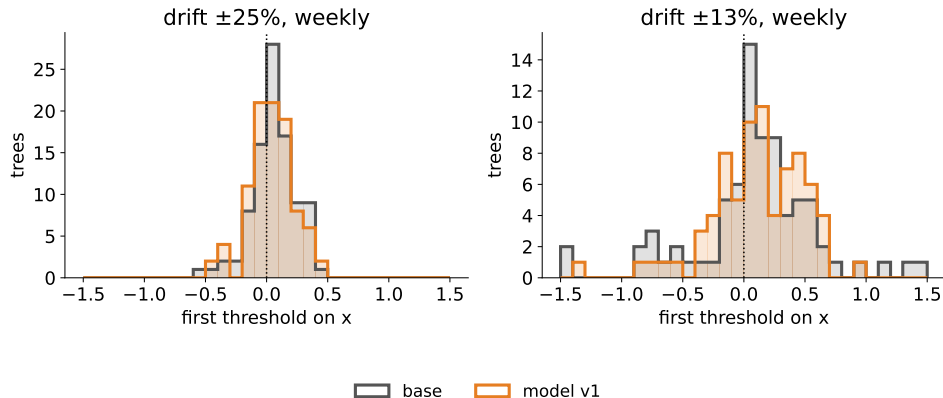
Model v1 against the base (E9)



Same 100 datasets for v1 and the base. Left: test Sharpe of v1 minus the base, dataset by dataset; mean, bars 2 s.e. Right: datasets keeping a second-level split, drift $\pm 25\%$.

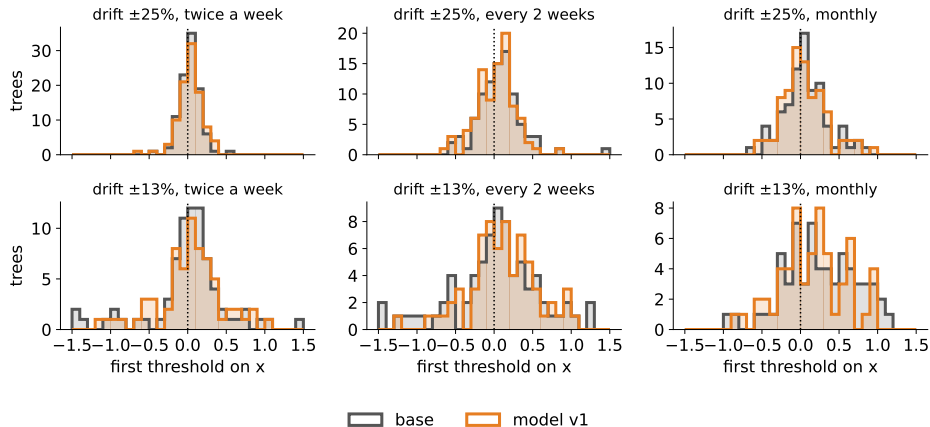
Same Sharpe from weekly on; spurious second-level leaves gone; smaller trees. **One cost:** 2 days, -0.08 (± 0.07). **Verdict:** cleaner structure, not a higher Sharpe. Full table: Appendix C.

E9: the distribution of the first threshold, v_1 against the base



Same 100 datasets per panel; weekly; $\lambda = 1$; 3 bp. First threshold of every tree that splits on x ; planted value 0. Orange: v_1 ; grey: base.

E9: the first threshold at the other decision rates, v1 against the base



E9, $\lambda = 1$, 3 bp, 100 datasets per panel. First threshold of every tree that splits on x; planted value 0. Orange: v1; grey: base; weekly: previous frame. Trees splitting on x: 81 to 99 of 100 at $\pm 25\%$, 59 to 75 at $\pm 13\%$, v1 and base alike.

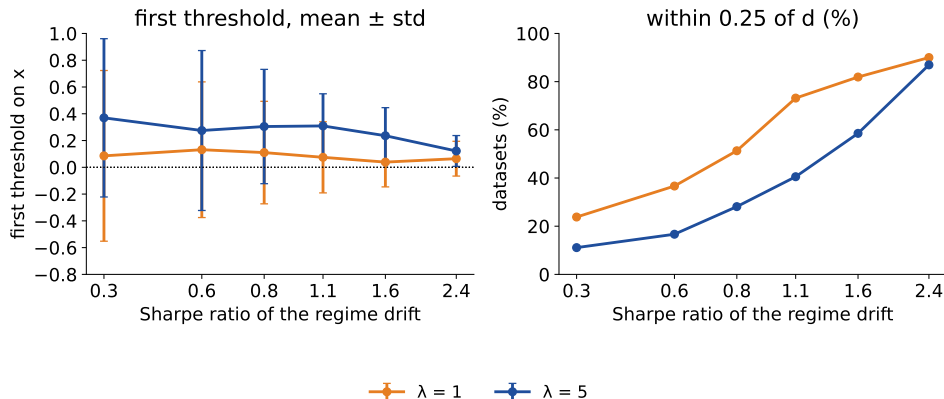
What can be found at all: resolution against volatility and feature noise (E6)

sweep (weekly)	value	edge	median $ \theta $	tree / ideal
σ per day (a year)	0.4% (6%)	0.25	0.05	1.85 / 2.10
	1% (16%)	0.10	0.12	0.69 / 0.87
	2% (32%)	0.05	0.39	0.23 / 0.42
	4% (63%)	0.025	0.95	-0.05 / 0.18
$x + \tau \varepsilon$ observed	0	0.10	0.12	0.69 / 0.87
	0.25	0.10	0.19	0.66 / 0.79
	0.5	0.10	0.25	0.45 / 0.66
	1	0.10	0.55	0.26 / 0.48
	2	0.10	1.15	0.09 / 0.29

60 datasets per row. Volatility sweep at drift $\pm 25\%/yr$. $|\theta|$: distance of the first threshold from the planted 0. $\varepsilon \sim \mathcal{N}(0, 1)$ daily noise, τ its std in std of x ; the price follows the true x . Tree / ideal: test Sharpe.

- threshold error $\approx 0.012/e$ down to $e \approx 0.05$; below: a guess
- noise τ on $x \approx$ edge divided by $1 + \tau^2$
- $\tau \leq 0.25$: free; 0.5: error doubles; 1: ideal Sharpe halves
- edge below a few bp/day: unlearnable from 15 years, by any method
- **Pre-test of a real feature:** measure its half-life and its noise τ ; estimate the regime Sharpe it can carry; read these curves and E10: regime Sharpe 0.3 = a guess, 0.8 = found one time in two, 1.6 = four in five

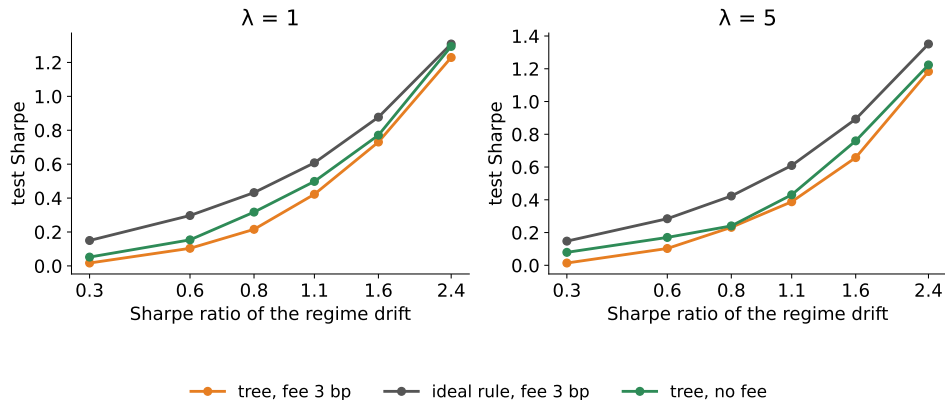
E10: the planted threshold against the strength of the signal



Model v1; weekly; 3 bp; volatility 16%; half-life 35 days. Drift ± 5 to $\pm 38\%$ a year = regime Sharpe 0.3 to 2.4; 50 datasets per point (100 at 0.8 and 1.6). Left: first threshold, mean $\pm$ one std over the trees that split; planted value 0. Right: first threshold within 0.25 of it.

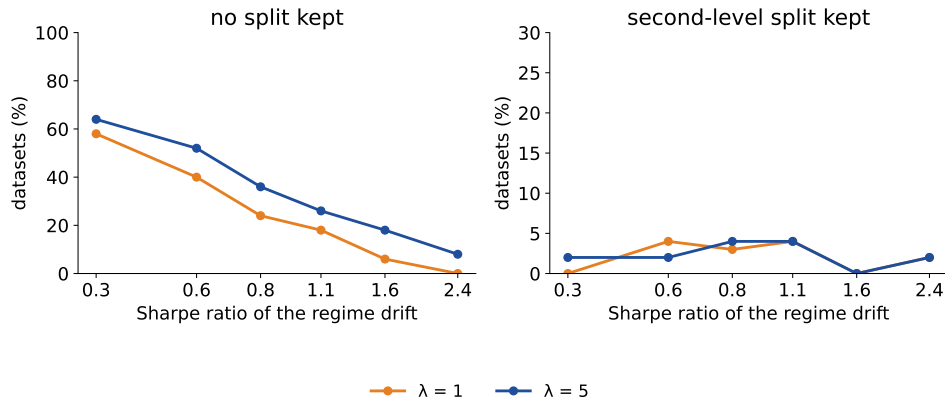
$\lambda = 5$ sits above 0 on purpose. Ideal position: 0 for $x < 0$, then a ramp, 0.14 at $x = 0$, 0.55 at 0.2, 0.84 at 0.5, 0.9 from 0.75 on (position = expected return / κ , clipped at 0 and 1). One cut with two answers fits that ramp best in its middle, not at 0.

E10: the Sharpe ratio against the strength of the signal, with and without fees



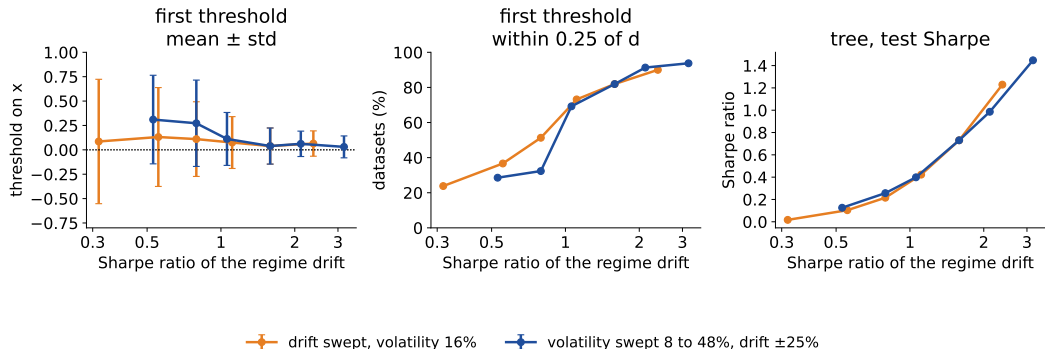
Same sweep. Test Sharpe: tree with 3 bp, tree without fees, ideal rule (optimizer fed the true expected return); $\lambda = 1$ and $\lambda = 5$.

E10: trees that keep no split, trees that keep a second-level split



Same sweep, 3 bp. Left: pruning leaves a single leaf. Right: a split below the root survives; one threshold in the market, so spurious.

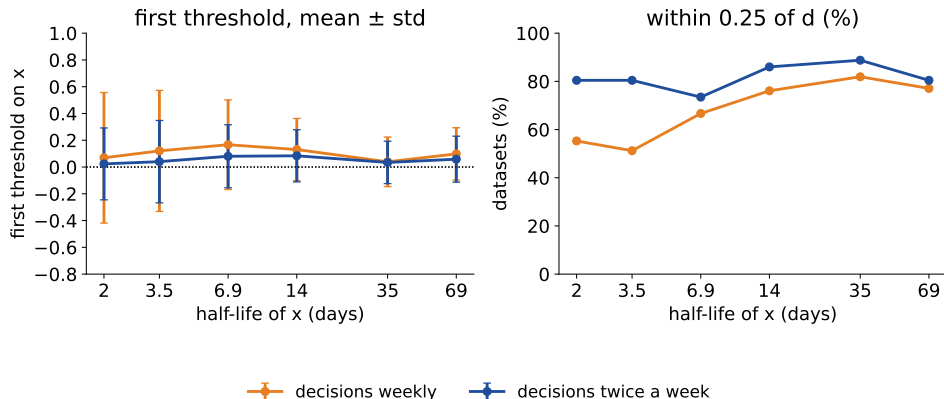
E10: does only the Sharpe ratio matter? Volatility sweep against drift sweep



Model v1, $\lambda = 1$, weekly, 3 bp, half-life 35 days, 50 datasets per point. Orange: drift raised at 16% volatility (previous frames).

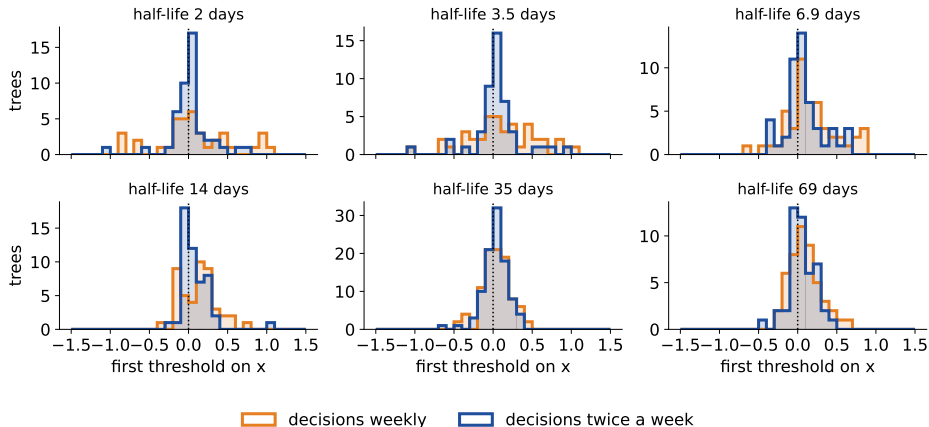
Blue: volatility raised from 8 to 48% at drift $\pm 25\%$; same Sharpe axis, 3.2 (8%) to 0.5 (48%). **If only the Sharpe ratio counted, the curves would coincide.** They do down to 24% volatility. At 32–48%: blue below; the risk term $\kappa = 2\lambda\sigma^2$ as large as the drift, ideal position a ramp, threshold shifts up, as at $\lambda = 5$.

E10: the half-life of the feature: the threshold



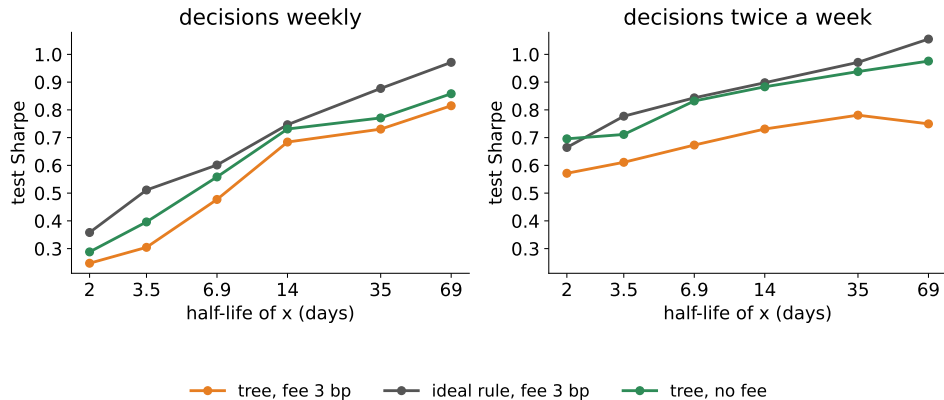
Model v1, $\lambda = 1$, 3 bp, drift $\pm 25\%$, volatility 16%, 50 datasets per point. Half-life of x 2 to 69 days (35 elsewhere in this deck); weekly or twice a week.

E10: the distribution of the first threshold by half-life



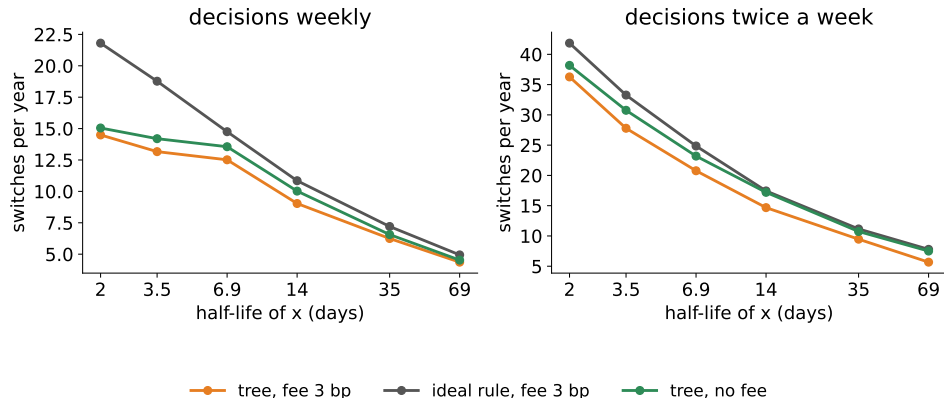
E10, $\lambda = 1$, 3 bp, drift $\pm 25\%$, volatility 16%, 50 datasets per half-life (100 at 35 days). First threshold of every tree that splits on x; planted value 0. Orange: weekly; blue: twice a week. Trees splitting: 38 to 48 of 50 weekly, 46 to 50 twice a week (94 and 98 of 100 at 35 days).

E10: the half-life of the feature: the Sharpe ratio



Same sweep. Test Sharpe: tree with 3 bp, tree without fees, ideal rule.

E10: the half-life of the feature: how often the tree switches



Same sweep. Full moves asset $\leftrightarrow$ cash per year on the test years, tree and ideal rule. **Fee bill in Sharpe units:** one full move = 0.03% of the portfolio; 36 a year = 1.1% a year; in the asset half the time, so volatility $\approx 11\%$: $1.1/11 \approx 0.10$ of Sharpe.

E10 in six points

- **The regime Sharpe sets the precision.** Within 0.25 of d : $\pm 25\%$ in 8 datasets of 10, $\pm 5\%$ in 1 of 4, and then half the trees keep no split. Fees: almost nothing.
- **Volatility acts through that Sharpe**, up to 32–48% a year: there the risk term makes the ideal position gradual and the recovery drops below what the Sharpe alone predicts.
- $\lambda = 5$: **the split sits above d , on purpose.** Ideal position = 0 below d , a ramp up to 1 above; the best single cut is in the middle of the ramp.
- **Second splits are gone.** At most 8% of the datasets keep one.
- **A fast feature needs a fast decision rate; the fee bill is part of the price.** Half-life of a few days: weekly cannot follow, the tree switches less than the ideal rule. Twice a week: threshold and most of the ideal Sharpe back, at 21–36 moves a year = 0.6–1.1% a year = 0.06–0.10 of Sharpe: nearly the whole gap to the no-fee tree at half-lives 2–4 days, less than half from 14 days on.
- **The rest of the gap: the fee inside the training freezes one tree in five at twice a week.** Leaf score within 0.03% of zero = inside the no-trade band: the tree holds a drifted mix for years, 0.2 instead of 0.9. No fee in the training: none freezes. Hence v2, item 1.

E11: the second market: a threshold for the drift and one for the volatility

Same feature; the price now has two drift regimes and two volatility regimes (t in trading days):

$$dx_t = k(m - x_t) dt + s\sqrt{2k} dW_t$$

$$\frac{dS_t}{S_t} = \mu_t dt + \sigma_t dB_t, \quad \mu_t = \begin{cases} +\mu & x_t \geq d \\ -\mu & x_t < d \end{cases}, \quad \sigma_t = \begin{cases} \rho\sigma & x_t \geq d_\sigma \\ \sigma & x_t < d_\sigma \end{cases}$$

$\sigma = 1\%$ a day (16% a year), $\rho = 0.5$ (calm regime: half the volatility), $e = 0.10$ (drift $\pm 25\%$ a year), $d = 0$; rest as in the first market.

Two placements of the volatility threshold

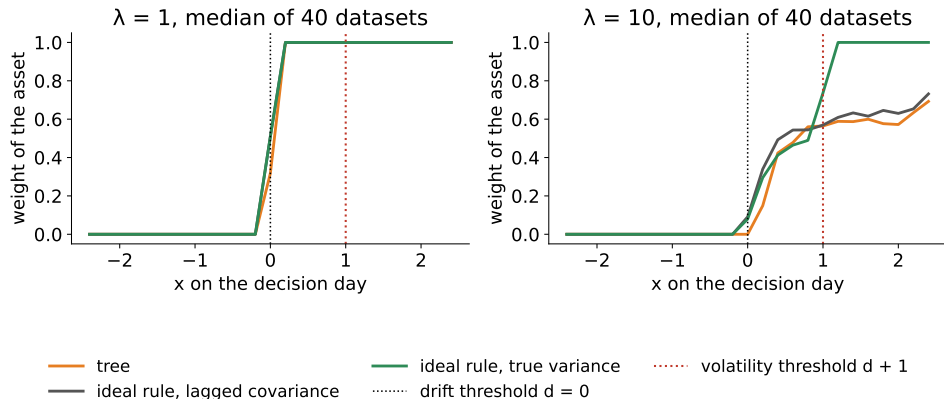
- $d_\sigma = d$: bad regime = noisy regime, the usual picture of markets
- $d_\sigma = d + 1$: good regime split in two: stressed $0 \leq x < 1$, calm $x \geq 1$

What a correct tree should do

- expected return independent of the volatility: drift threshold stays at $d = 0$
- volatility acts through the position, $w = (g_h(x) \mp f)/(2\lambda\sigma_h^2)$ clipped to $[0, 1]$: $\lambda = 1$, all or nothing in both regimes, threshold invisible; $\lambda = 10$, one half in the stressed part, one in the calm part: a second boundary at $x = 1$, of size not sign

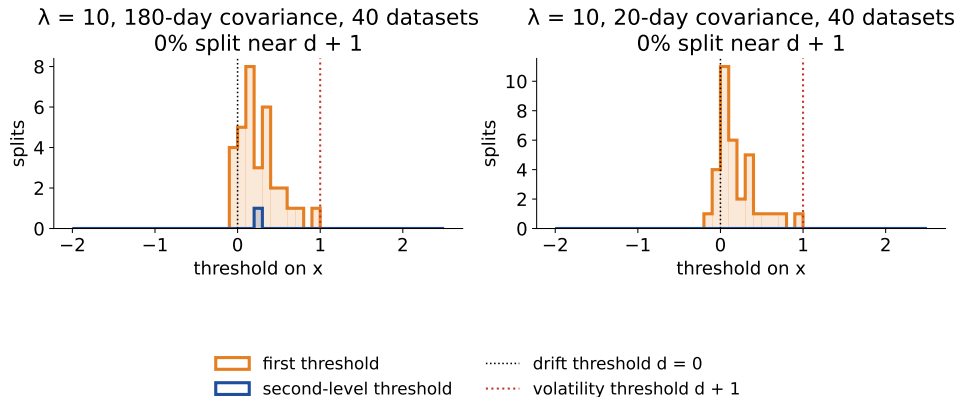
Seven settings: constant volatility, $d_\sigma = d$, $d_\sigma = d + 1$, at $\lambda = 1$ and 10, plus $d_\sigma = d + 1$, $\lambda = 10$, 20-day covariance; rates 2, 5, 10, 21 days; drifts ± 13 , $\pm 25\%$; 40 datasets per cell: 2,240 fits. Base: E7 (23 September); model v1: E11 (28 September).

E11: the second market on model v1: the position taken by x



E11: volatility 1% a day below $x = 1$, 0.5% above; drift threshold 0; weekly; drift $\pm 25\%$; 40 datasets; median position by x on the test decisions. $\lambda = 1$: 0 or 1 whatever the variance. $\lambda = 10$: ideal = one half in the stressed part of the good regime, one in the calm part; the tree follows the ideal rule with the same lagged covariance, not the one that knows the true variance.

E11: the second market on model v1: no split at the volatility threshold



Same setting, $\lambda = 10$: every threshold kept by the 40 trees, 180-day covariance and 20-day covariance. v1: no split near the volatility threshold (the base: 8%). The variance boundary is invisible to the regret, same lagged covariance as the optimizer; the hurdle removes the rest.

A fair assessment, and what comes next

Holds

- $\pm 25\%$ regime, monthly feature: recovered at every rate; 75–85% of the attainable gain
- $\pm 13\%$: graceful; $\pm 5\%$: correctly nothing
- v1 = base with smaller trees, steadier thresholds (100 paired datasets); cost 0.08 at 2 days
- second market (E7, E11): drift threshold found better when the calm regime pays, Sharpes nearly double; a variance-only boundary never learnt (lagged covariance in the regret)

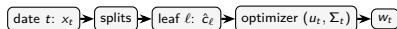
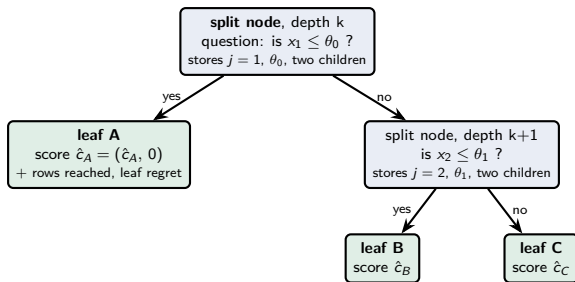
Weak or untested

- no option beats the base beyond noise: structure, not Sharpe
- growth Sharpe gate: in-sample, mixes a risk preference into the structure
- one clean feature, synthetic data only, decoys only at an unrealistic edge, depth budget unused

Next (v2, decided on 25–28 September)

- fees out of the score-search regret, kept in the Sharpe gate and the pruning; evidence: the freeze of E10
- candidate thresholds at the quantiles of their distribution; rescore to a fixed point; exact leaf scores
- each change: an option off by default, tested alone and combined on the 100 datasets of E9, paired with v1; documented in the same step
- still open: growth Sharpe gate off; regime-aware variance; two features + decoys at market sizes; real data, each feature pre-tested as on the E6 frame

The tree: what a node stores



- tree = nested questions “is $x_j \leq \theta$?”, ending in leaves
- **split node**: feature j , threshold θ , two children; no score
- **leaf**: one score vector $\hat{c}_\ell = \left(\hat{c}_\ell^{(1)}, 0 \right)$ (only the difference matters); rows reached; leaf regret
- holdings u_t , weights w_t , returns: per date, not per node
- depth $\leq 3 \Rightarrow \leq 8$ leaves $\Rightarrow \leq 8$ scores
- growth: leaf $\rightarrow$ split node + 2 leaves; pruning: the reverse
- use: $x_t \rightarrow$ leaf $\rightarrow \hat{c}_\ell \rightarrow$ optimizer $\rightarrow w_t$

The decision: mean–variance with proportional fees

Notation

- u_t : holdings between time $(t-1)^+$ and t^- ,
 $u_{t+1} = \frac{w_t \circ (1 + r_t)}{1 + r_t^\top w_t}$, $u_1 = \mathbf{1}/n$
- c : the tree's scores, in units of the return; cash = 0 (only the score differences matter)
- Σ : covariance of the h -day return = $h \times$ the covariance of the last 180 daily returns; $\sigma_h^2 = h\hat{\sigma}^2$
- f : fee per traded notional, 3 bp on the asset, 0 on cash; $\lambda = 1$: risk aversion, weight of the h -day variance
- $\mathcal{W} = \{w : 0 \leq w_i \leq \bar{w}, \sum_i w_i = 1\}$

Properties

- strictly concave $\Rightarrow$ unique w^*
- $w^*(c + \alpha \mathbf{1}) = w^*(c)$: only score differences matter
- exact solution: 5 states per weight (0, cap = $\bar{w}$, u_i , buy, sell) $\Rightarrow 5^n$ linear systems, best feasible one

$$w_t^*(c; u_t, \Sigma) = \arg \max_{w \in \mathcal{W}} \underbrace{c^\top w}_{\text{gain}} - \underbrace{f^\top |w - u_t|}_{\text{fees}} - \underbrace{\lambda w^\top \Sigma w}_{\text{risk}}$$

Asset against cash: the position window and the no-trade band

One risky asset (weight w , score c , weight held before the decision u) against cash (score 0, no fee, no variance); $\kappa = 2\lambda\sigma_h^2$:

$$w^* = \arg \max_{0 \leq w \leq 1} c w - f |w - u| - \frac{\kappa}{2} w^2$$

Without fee ($f = 0$): $w^* = c/\kappa$, clipped to $[0, \bar{w}]$.

With a fee f per traded notional: moving away from u costs $f|w - u|$, so the score must beat the fee before anything moves:

$$w^* = \begin{cases} \min\{(c - f)/\kappa, \bar{w}\} & c - \kappa u > f \text{ (buy)} \\ \max\{(c + f)/\kappa, 0\} & c - \kappa u < -f \text{ (sell)} \\ u & |c - \kappa u| \leq f \text{ (no trade)} \end{cases}$$

- the position is linear in the score inside the *position window* $0 \leq c \leq \bar{w}\kappa$: all cash below 0, fully invested above κ
- $\lambda = 1$, weekly, $\sigma = 1\%$ a day:
 $\kappa = 2 \cdot 1 \cdot 5 \cdot (1\%)^2 = 0.1\%$ per 5 days

The no-trade band is $|c - \kappa u| \leq f$: half-width $f = 3$ bp around κu , the score that would justify the position already held. A score inside it leaves the portfolio as it is, whatever it holds.

decision rate	window κ	band half-width f	band / window
weekly ($h = 5$)	0.10%	0.03%	30%
twice a week ($h = 2$)	0.04%	0.03%	75%

v1: leaf scores of ± 0.3 – 0.5% per 5 days sit far outside the window $\Rightarrow w \in \{0, 1\}$; continuous sizing needs $\lambda \geq 5$. But a score near 0 with a drifted u sits inside the band, and the tree then holds that mix (E10).

The loss: decision regret; the score of a leaf

$$U_t(w) = r_t^\top w - f^\top |w - u_t| - \lambda w^\top \Sigma_t w$$

$$\text{Reg}_t(c) = U_t(w^*(r_t; u_t, \Sigma_t)) - U_t(w^*(c; u_t, \Sigma_t)) \geq 0$$

hindsight decision – decision on c (given the same holdings u_t)

$$\hat{c}_\ell = \arg \min_{c \in [-b, b]^n} \sum_{t \in \ell} \text{Reg}_t(c), \quad b = 10\% \text{ per holding period, needed for bound + resolution}$$

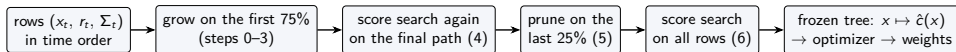
Leaf score

- one regret per date (r_t depends on t), summed over the leaf's dates
- bounded coordinate search: 3 passes $\times$ 7 values
- not a forecast: same regret for any score on the right side

Consequences

- fees $\Rightarrow$ regret depends on $u_t \Rightarrow$ holdings are frozen for the round (no backpropagation) then replayed (drifted)
- only differences: relative returns, asset vs cash $\Rightarrow$ help to choose the right features
- same Σ_t as the optimizer: unseen risk not learnt

Training: replay, two growth gates, out of sample pruning



0. root score from $u_t = 1/n$; **replay** (trade the scores date by date from u_1) $\Rightarrow u_t$, Sharpe S ; holdings frozen for the round
1. **regret gate**, per leaf: score search again from current u_t ; candidates; children of the 3 best; keep iff $\Delta/N_\ell > 10^{-6}$, $\Delta = L_\ell - L_{\text{left}} - L_{\text{right}}$ (regret per date, in h -day return)
2. **Sharpe gate**: rank all kept splits by Δ ; whole-tree replay of the first; accept iff $S' > S$, else next; one per round; none $\Rightarrow$ stop
3. replay the new tree $\Rightarrow$ new u_t ; back to 1 (depth ≤ 3 ; ≥ 20 rows and $\geq 10\%$ of the parent per leaf)
4. score search again on every leaf, final path; kept iff S does not fall
5. **pruning**, bottom-up, on the last 25% (never used for growth): with vs without the split, both replayed; $g_t = \text{Reg}_t(\text{without}) - \text{Reg}_t(\text{with})$; keep iff $\bar{g} > 10^{-6}$, $\bar{g} > k_p \text{se}(\bar{g})$ [$k_p = 2$ below the root, 0 at the root], checking Sharpe $\uparrow$
6. score search again on all training rows

Gates 1–2: in-sample. Gate 5: out-of-sample.

Split-search options: quantile candidates, refinement, soft splits

Common: admissible candidates $\theta_{(1)} < \dots < \theta_{(K)}$ (each side $\geq m_* = \max(m, \lceil \varphi N \rceil)$ rows; $m = 20$, $\varphi = 10\%$); cheap screen

$$Q(\theta) = \sum_{t \in I} \text{Reg}_t(c_{\text{side}(t)}(\theta)), \quad c_{\text{left}}(\theta) = \text{clip}\left(\text{mean}\{r_t : x_t \leq \theta\}, \pm b\right), \text{ cf Theorem 1, idem right}$$

Deciles by rank + refinement

- keep ranks $\text{round}(q(K-1))$, $q = 0.1, \dots, 0.9$: 9 screens instead of $K \approx N - 2m_*$
- refinement: screen every candidate between the two deciles around the best one, θ_o ; best of them $\theta_\bullet$
- replace iff $Q(\theta_o) - Q(\theta_\bullet) > k \text{ se}$, $\text{se} = \sqrt{|I|} \text{std}_t(\delta_t)$, $k = 1$
- 3 best survivors ranked by Q ; replay the first, then the second, then the third; keep iff Sharpe $\uparrow$

Soft split (off in v1: $\kappa = 0$)

$$z_k = \frac{Q(\theta_k) - \min_I Q(\theta_I)}{\text{se}_k}, \quad \pi_k = \frac{e^{-z_k/\kappa}}{\sum_I e^{-z_I/\kappa}}$$

- all candidates kept, weighted; Gaussian variant $e^{-z^2/2\kappa^2}$
- membership: left with $\mu(x) = \sum_{k: \theta_k \geq x} \pi_k$, right with $1 - \mu(x)$; weights $\omega_{t\ell}$ multiply down the tree
- leaf fit: $\min_c \sum_t \omega_{t\ell} \text{Reg}_t(c)$; row score $\hat{c}(x_t) = \sum_\ell \omega_{t\ell} \hat{c}_\ell$: a ramp, not a jump
- $\kappa \rightarrow 0$: plain split

Model v1: the frozen configuration (24 September 2026)

Choice	v1	Why (section of the experiments report)
risk aversion λ	1 (weight of the h -day variance)	positions 0/1: regime classifier, not sizer (8)
fee f	3 bp = 0.03% of the traded notional on the asset; 0 on cash	market-sized; 10 bp starves fast rates (8)
decision rate h	weekly ($h = 5$ trading days); every 2 days as alternative	daily $\approx$ weekly; monthly loses (4)
depth	≤ 3	depth alone changes nothing (8)
leaf minimum	20 decisions and 10% of the node	tail leaves at 0.7% of the data (7)
candidate thresholds	deciles by rank + refinement (1 s.e.)	= base Sharpe, steadier thresholds (9, 12)
soft splits	off	never raise the Sharpe at $\lambda = 1$ (9, 12)
growth gates	regret reduction, then replayed Sharpe $\uparrow$	kept as designed; v2 revisits
pruning	last 25%; hurdle 2 s.e. below the root, none at the root	out-of-sample judge (7); removes spurious leaves at no cost (12)
covariance	$\Sigma_t = h\hat{\sigma}_t^2, \hat{\sigma}_t^2$ the variance of the last 180 daily returns	20-day window: noisier than timely (regime report)
conventions	cash column; all-cash Sharpe = 0; $ \hat{c} \leq 10\%$ per holding period	

Base of every experiment = same pipeline, every option off: depth 2, all candidates screened, plain pruning.

Result 1: the threshold is recovered at market sizes (E2, base pipeline)

edge	h	split on x (%)	threshold within 0.25	annual. test Sharpe		
				tree	ideal	gain
0.10	1	97	95%	0.80	1.03	76%
	5	94	81%	0.73	0.88	79%
	21	81	60%	0.54	0.68	75%
0.05	1	82	60%	0.12	0.49	25%
	5	78	53%	0.24	0.43	51%
	21	62	34%	0.17	0.35	43%
0.02	5	54	26%	-0.03	0.15	-3%

- 100 datasets per row (50 at edge 0.02); planted threshold 0; edge 0.10, 0.05, 0.02 = $\pm 25\%$, $\pm 13\%$, $\pm 5\%$ a year; h in trading days
- split on x : trees keeping a split
- within 0.25: first threshold within 0.25 of 0
- Sharpe: mean over datasets, s.e. ≈ 0.04
- gain: tree's excess test return over the no-split tree / the ideal rule's

- $\pm 25\%/yr$: **found at every rate**; 95% within 0.25 daily, 60% monthly; 75–79% of the ideal gain
- $\pm 13\%/yr$: **graceful**; 62–82% split; 25–51% of a small gain (ideal 0.35–0.49)
- $\pm 5\%/yr$: **nothing to find in 15 years**; half no split, half at random; ideal rule ≈ 0.15 ; finding nothing = right answer
- daily $\approx$ weekly; monthly loses (regime changes inside the holding period)

Result 2: the strategy's performance out of sample (E2, base pipeline)

edge	h	annualized test Sharpe ratio				gain captured
		tree	ideal rule	rule at d	buy & hold	
0.10	1	0.80	1.03	1.04	-0.02	76%
	2	0.86	0.97	0.98	-0.02	85%
	5	0.73	0.88	0.88	-0.02	79%
	10	0.63	0.78	0.78	-0.02	78%
	21	0.54	0.68	0.66	-0.02	75%
0.05	1	0.12	0.49	0.49	-0.02	25%
	2	0.21	0.46	0.46	-0.02	44%
	5	0.24	0.43	0.42	-0.02	51%
	10	0.17	0.39	0.38	-0.02	43%
	21	0.17	0.35	0.34	-0.02	43%

- 100 datasets per row; s.e. ≈ 0.04
- ideal rule: optimizer fed $\mathbb{E}[r \mid x]$; rule at d : asset iff $x \geq d$; buy & hold: always the asset
- turnover per weekly decision (edge 0.10, both legs): tree 0.25, ideal 0.29 $\Rightarrow$ fees $\approx 0.2\%$ a year

- **Most of the signal captured:** $\pm 25\%$ regime, about three quarters of the ideal gain.
- **The threshold is the whole job.** Rule at d = ideal rule: at $\lambda = 1$ nothing else to learn.
- **Lost: precision.** Gap to the ideal = threshold error + a few spurious leaves.
- **A weak regime: a limit of the data, not of the tree.** $\pm 13\%$: part of a small gain; 15 years barely identify it.
- **No signal, no gain.** Buy & hold earns zero; all the gain is in the switch.

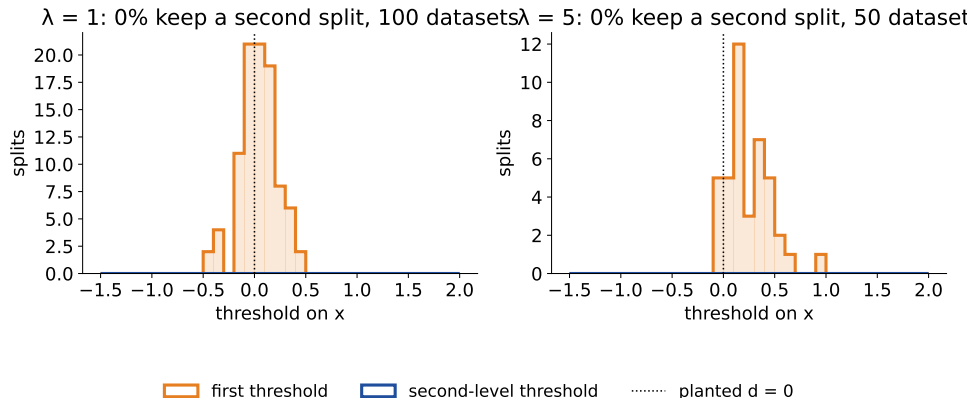
Model v1 against the base, on the same 100 datasets (E9, 800 fits)

edge	h	leaves		2nd level	within 0.25	Sharpe		v1 – base (2 s.e.)
		base	v1	base → v1	base → v1	base	v1	
0.10	2	2.24	2.00	25 → 2%	92 → 89%	0.86	0.78	−0.08 (0.07)
	5	2.14	1.94	18 → 0%	81 → 82%	0.73	0.73	+0.00 (0.04)
	10	2.03	1.92	14 → 2%	70 → 72%	0.63	0.64	+0.01 (0.05)
	21	1.90	1.81	9 → 0%	60 → 64%	0.54	0.53	−0.01 (0.03)
0.05	2	2.18	1.72	38 → 4%	64 → 57%	0.21	0.18	−0.03 (0.06)
	5	2.01	1.80	21 → 3%	53 → 51%	0.24	0.22	−0.03 (0.05)
	10	1.93	1.69	19 → 1%	47 → 49%	0.17	0.20	+0.03 (0.04)
	21	1.76	1.59	13 → 0%	34 → 42%	0.17	0.14	−0.03 (0.03)

same 100 datasets; Sharpe: test years, annualized; 2nd level: datasets keeping a second-level split; last column: mean paired difference, 2 s.e. in brackets; half of the datasets: identical Sharpe

- **The question:** does v1 beat the base it was built from? Same 100 histories: every difference paired.
- **Same Sharpe** from weekly on, within noise.
- **Cleaner trees:** spurious second-level leaves gone, threshold found as often, smaller trees.
- **One cost:** every 2 days, some second-level splits are real and the hurdle removes them.
- **Verdict:** v1 is the base with a cleaner structure, not a higher Sharpe.

E10: where the second-level thresholds land



Model v1, weekly, drift $\pm 25\%$, volatility 16%, 3 bp. Every threshold kept: first in orange, second-level in blue. None: v1's hurdle removes every split below the root in this market.